MIL-G-36294A
28 May 1969
SUPERSEDING
MIL-G-36294(DM)
30 September 1964

MILITARY SPECIFICATION

GRINDING AND POLISHING MACHINE, DENTAL LABORATORY, BENCH MOUNTED, HIGH SPEED, 115 VOLT, 60 CYCLE, AC

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high speed dental lathe suitable for grinding and polishing dental appliances fabricated of chrome-cobalt alloys, and rapid finishing of gold alloys.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

CC-M-636	Motor, Alternating-Current, (Fractional Horsepower).
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055	Barrier Material, Waterproofed, Flexible.

Military

MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold-Application.

FSC 6520

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES OF AMERICA (USA) STANDARDS INSTITUTE

USAS C73.11-1966	Outlet Receptacles, Attachment Plug Caps and Appliance Plugs

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, N.Y. 10016.)

UNDERWRITERS' LABORATORIES, INC.
Underwriters' Laboratories Standards

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L.I., N.Y. 11749, 207 E. Ohio Street, Chicago, Illinois, 60611; and 1655 Scott Blvd, Santa Clara, Calif, 95050.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

* 3.1.1 The contractor shall submit proof to the contracting agency, prior to delivery, that the equipment he will supply under this specification conforms to the requirements of Underwriters' Laboratories, Inc. Standards. The label or listing of the Underwriters' Laboratories, Inc. may be accepted as evidence that the equipment conforms to this requirement.

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3.1.2 In lieu of the label or listing, the contractor may submit independent proof satisfactory to the contracting agency, prior to delivery, that his equipment conforms to the Underwriters' Laboratories, Inc. Standards.

3.1.3 Compliance with Underwriters' Laboratories, Inc. requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the other requirements of this specification in order to secure the acceptance of his commodity.

* 3.2 Materials. Materials shall be new and shall conform to the applicable specifications and requirements hereinafter specified. Materials not specifically designated shall be of high commercial quality throughout, and shall be entirely suitable for the purpose intended.

* 3.3 Design. The high speed lathe shall be designed for operation from a 120 volt, 60 cycle, alternating current supply. The lathe shall be equipped with a high speed automatic spindle which shall operate at speeds of 10,000 revolutions per minute (r.p.m.) and 24,000 r.p.m. and shall be pulley driven by means of a 3 "VEE" belt and a 1/3 Horse Power (H.P.)continuous duty motor. An adjustable lighting fixture and safety glass shield shall adequately illuminate the work area and provide operator protection. Spindle clearance, confined by the work surface below the spindle, splash guard behind the spindle, and lighting fixture and shield above the spindle, shall be adequate to facilitate manual manipulation of dental appliances during grinding and polishing operations. The high speed lathe shall be of heavy duty design and shall be suitable for grinding and polishing dental appliances fabricated of chrome-cobalt alloys and for rapid finishing of gold alloys. The lathe shall be constructed to accept a vacuum dust collector which may be attached if desired (not supplied with this item). The dust collector shall be available as an accessory.

* 3.4 Construction. Except for the motor and automatic spindle, structural members shall be fabricated of cast aluminum. All members shall have cross sectional areas of sufficient magnitude and shall be adequately fastened to prevent resonance and acoustical vibrations. Hardware in contact with aluminum surfaces shall be fabricated of the same material or of 18-8 PH stainless steel. As an alternate, steel suitably plated to prevent galvanic corrosion shall be used. The weight of the grinding and polishing machine shall be 30 ± 5 pounds and the overall dimensions shall be 13 ± 3 inches wide by 15 ± 3 inches deep by 13 ± 3 inches high. Essentially, the grinding and polishing machine shall consist of a base, motor, splash guard, lighting fixture, safety glass shield, automatic spindle, pulley drive, and associated electrical components.

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3.4.1 Base. The base shall be fabricated of cast aluminum with integral work surface and mounting pads. The work surface shall be located below the top surface of the base forming a cavity open on the operator's side, and shall be of sufficient depth and area to be suitable for the purpose intended. The mounting pads shall be provided with heavy rubber feet to prevent slipping or damage to the bench top.

* 3.4.2 Motor. The motor furnished shall be in accordance with CC-M-636, for type 1, class 1b, style A1, or for type III, class 7, style A1, and shall be rated 1/3 H.P., 3450 r.p.m., continuous duty, for operation from a 120 volt, 60 cycle, alternating current supply.

3.4.3 Splash guard. A splash guard shall be provided between the motor and spindle and shall be of sufficient height and width to shield the motor.

3.4.4 Lighting fixture and safety glass shield. The lighting fixture and safety glass shield shall be an integral unit and shall be adjustable to illuminate the work area, reduce shadows, and accommodate viewing angle. The light bulb receptacle, light switch, and safety glass shield shall be securely mounted to the lighting fixture housing.

3.4.5 Automatic spindle. The automatic spindle shall be designed for continuous duty and shall operate at speeds of 10,000 r.p.m. and 24,000 r.p.m. Integral components of the automatic spindle shall include a draw type collet, sealed pre-lubricated precision bearings, cast aluminum housing, and quick change mechanism with actuating lever. The collect shall be fabricated of carbon tool steel, low-tungsten steel, or a chrome-vanadium alloy steel and shall have a 3/32-inch arbor capacity. The draw end and shank of the collet shall have a Rockwell hardness of C37 to C41, and the collet hole shall be hardened to Rockwell C60 minimum. The quick change mechanism shall be lever-operated and shall provide an efficient means whereby the spindle is stopped and collet pressure relieved without disengaging the pulley drive or stopping the motor. Materials and associated hardness requirements for component parts of the spindle shall be selected in accordance with good engineering practice, with wear characteristics favoring those components that are readily replaceable.

* 3.4.6 Pulley drive. A dual speed pulley drive shall be provided between the motor and automatic spindle designed to satisfy spindle speed requirements of 3.4.5. Speed changes shall be obtained by manually repositioning the drive belt to change pulley ratios. A removeable vacuum formed plastic belt guard shall be provided to protect the operator and to facilitate speed change requirements.

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* 3.4.7 Electric wiring. Separate circuits and control switches shall be provided for the motor and for the lighting fixture. A 3-conductor, Number 18 gage type SJ power cord 3-1/2 feet minimum length shall be supplied with a 3-prong plug on one end suitable for use with a 125 volt, 15 ampere, 2 pole, 3 wire grounding type receptacle as specified in the United States of America Standards Institute, USA Standard C73.1 - 1966. The lighting fixture wire provided shall be type SPT-2 suitably routed and of sufficient length to accommodate adjustment of the lighting fixture. The power cord and fixture wire shall terminate in a suitable metal enclosure, to which the motor switch shall be mounted. All wire terminations and splices shall be suitably strain relieved and insulated against shorting.

* 3.4.8 Switches. Control switches shall be of the snap action toggle type construction, totally enclosed, and shall have a current carrying capacity of at least 125 percent of the maximum circuit requirements for 120 volts, 60 cycles, A.C. A cover plate shall be provided with each switch with "ON" and "OFF" switch positions legibly and permanently designated thereon.

3.5 Performance. Performance criteria shall be based upon operating characteristics as well as functional design. Normal operation of the grinding and polishing lathe shall require no special tools. The grinding and polishing lathe shall include the requirements specified in 3.5.1 through 3.5.4.

* 3.5.1 Spindle runout. Spindle runout shall not exceed 0.001 inch when measured 1/8 inch from the collet face, when tested as specified in 4.4.1.

3.5.2 Belt tension adjustment. The features provided for adjusting belt tension shall also prevent misalignment of components. The belt shall not walk off the pulleys nor shall the belt snap or whip to the extent that the equipment is considered unsuitable for use, when tested as specified in 4.4.2.

3.5.3 Pulley design. Pulley design shall be such that a belt shift during speed changes shall not result in performance specified in 3.5.2, when tested as specified in 4.4.3.

3.5.4 Quick change mechanism. Operation of the quick change mechanism shall result in a smooth transition from operating speed to full stop without squealing, chattering, or grinding, when tested as specified in 4.4.4.

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3.6 Finish. Aluminum castings shall be heat-treated and age-hardened to provide stability where positional accuracy must be maintained. Castings shall be completely deburred with mounting surfaces machined, and shall be furnished in a natural finish. Exposed components not fabricated of corrosion-resisting metal shall be suitably plated or rendered corrosion-resisting by suitable rust preventive measures.

3.7 Interchangeability. Replaceable parts shall be manufactured to necessary standards in order to permit replacement without modification of component parts or machine assemblies.

* 3.8 Identification marking. Each unit shall be provided with a readily visible nameplate permanently affixed thereto. The plate shall be marked with the name or registered trademark of the manufacturer, model designation, serial number, current, voltage, and frequency. The markings on the plate shall be clear and permanent, and shall be accomplished by die stamping. Alternate methods of marking the nameplate (other than decalcomania) may be submitted to the contracting officer for approval.

* 3.9 Service data. The contractor shall enclose, with each unit, two copies each, of one or more service data booklets. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. The service data shall contain complete instructions for installation, operation, maintenance, and a list of component parts. Wiring diagrams shall be included for electrical items or components. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number.

* 3.9.1 Approval. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order.

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* 3.9.2 Waiver of review and distribution copies of manual. Should the supplier have furnished manuals which have been reviewed, commented on, and approved by the Directorate of Medical Materiel for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.10 Workmanship. Workmanship shall be first class throughout. The grinding and polishing machine shall be free from defects which may detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

* 4.2 Quality conformance inspection. The examination and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I and II (see 4.2.1 and 4.2.2) and table III (see 4.2.3).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table I. Unit of product for examination purposes shall be one grinding and polishing machine.

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* 4.2.3 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one grinding and polishing machine.

Table III. Tests

Characteristic	Requirement	Test Procedure	Inspection Level
Spindle runout	3.5.1	4.4.1	S-2
Belt tension	3.5.2	4.4.2	S-2
Pulley design	3.5.3	4.4.3	S-2
Quick change mechanism	3.5.4	4.4.4	S-2

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Spindle runout. A 0.093 diameter drill rod shall be mounted in the collet and shall extend 3/16 of an inch beyond the collet face. A dial indicator of suitable sensitivity shall be used to measure maximum runout while manually rotating the spindle.

4.4.2 Belt tension adjustment. Completely relieve belt tension by means of adjusting features provided. Retighten belt and rotate pulleys through several revolutions and apply electrical power to the grinding and polishing machine and test for characteristics specified in 3.5.2.

4.4.3 Pulley design. To change pulley ratios, shift and drive belt by means of features provided. Apply power to the grinding and polishing machine and test for characteristic specified in 3.5.3.

4.4.4 Quick change mechanism. Operate the lathe at the required spindle speeds and, with the features provided, actuate the quick change mechanism to engage and disengage the 3/32 drill rod (see 3.5.4).

5. PREPARATION FOR DELIVERY

* 5.1 Packaging. Packaging shall be level A or C, as specified, (see 6.1).

5.1.1 Unit of issue. One grinding machine, as specified, constitutes one unit of issue.

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5.1.2 Level A. Unit preparation. Exterior base metal surfaces subject to corrosion shall be treated with corrosion preventive compound conforming to grade 3 of MIL-C-16173. All exterior finished surfaces shall be protected by wrapping with Kraft paper and cushioning with excelsior pads or flexible corrugated fiberboard or other suitable materials. Wrapping and cushioning material shall be secured by tying with cotton cord or tape, or by sealing with Kraft gummed paper tape. Work shall be accomplished in such a manner as to insure against marring, scratching, or damage to the unit.

* 5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

* 5.2 Packing. Packing shall be level A or C, as specified (see 6.1).

* 5.2.1 Level A. Each unit shall be overpacked in an exterior container designed for type 3 load, and constructed in accordance with PPP-B-601, table III, using type I, class 1 plywood, or PPP-B-621, table IV. The unit shall be adequately secured within the exterior container by means of bolts, wood braces, or other interior fittings, to insure against shifting or breakage. Each container shall be lined with waterproof barrier material conforming to PPP-B-1055, class L-2. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

* 5.2.2 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Exterior containers. Exterior containers shall be marked in accordance with MIL-STD-129. Serial numbers shall also be shown.

6. NOTES.

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

6.3 Specification changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities
Army - MD
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION: MIL-G-36294A FSN:

ORGANIZATION:

CITY AND STATE:

CONTRACT NUMBER:

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*:

DATE:

DD FORM 1426 REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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